MASTER IN ASTROPHYSICS AND SPACE SCIENCE

УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

Spectroscopy of Astrophysical Plasmas

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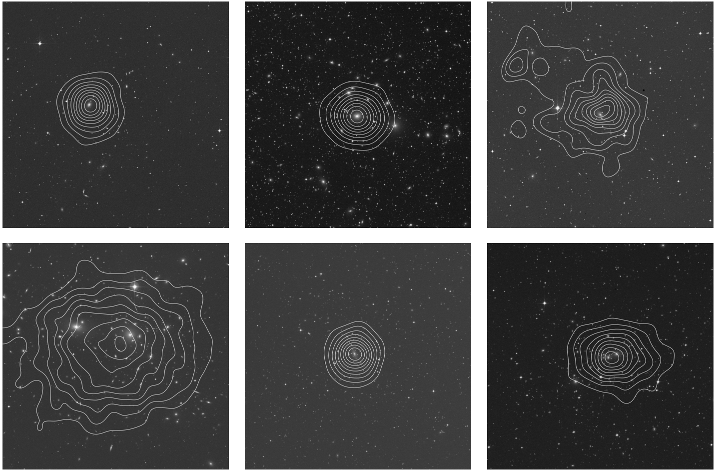


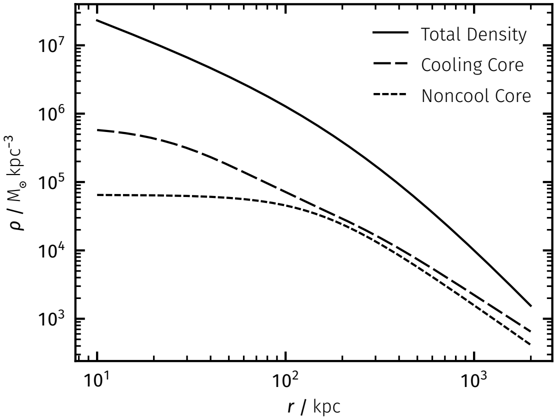
Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.

Overview

Overview

Cluster Models & Classes

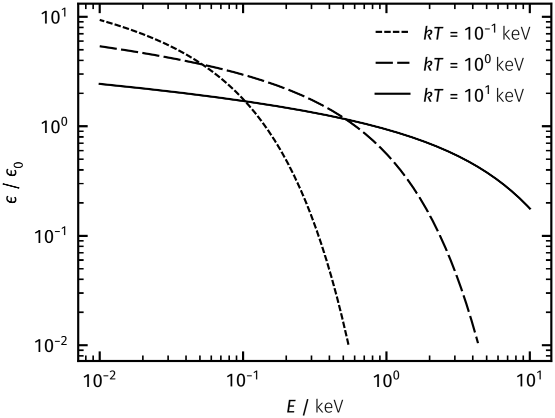
- Sparse & Hot Plasma Systems
- Collisional Ionization Equilibrium (CIE)
- Purely Gravitational Baseline Assumption
- Dark Matter Halo Dominated Objects
- Selfsimilar Radial Scaling
- Deviation Defines Categories
    - Cooling Cores (CC)
    - Noncool Cores (NCC)
- Mergers Inject Energetic Shocks & Sloshing
- Central Active Galactic Nuclei (AGN) Feedback



Overview

Characteristic Emission

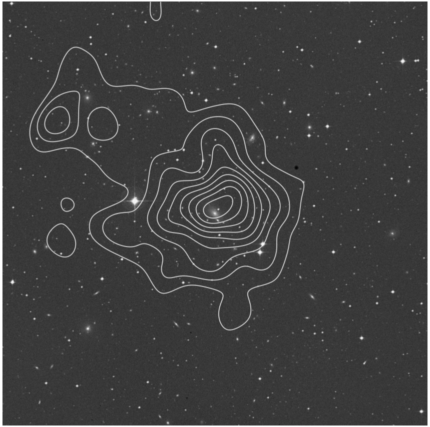
- Bremsstrahlung *(free-free)*
  - Broad Continuous Spectrum
- Recombination *(bound-free)*
  - Soft Continuum
  - Hard Ionization Edge
- Atomic Transitions *(bound-bound)*
  - Discrete Lines
- Sunyaev-Zeldovich Effect *(Inverse Compton)*
  - Cosmic Microwave Background Photons



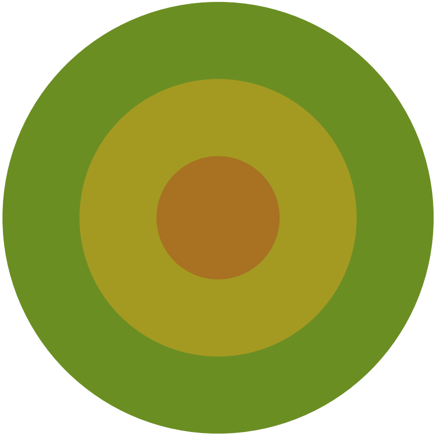
Overview

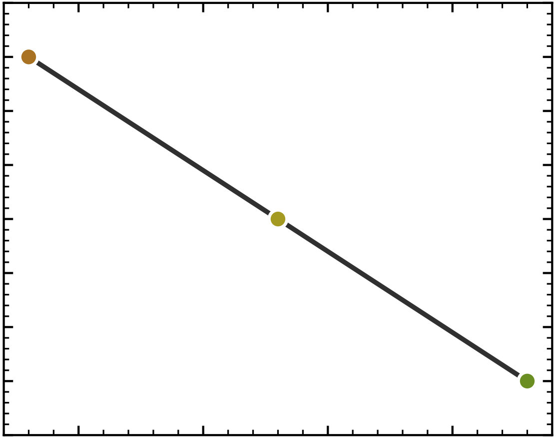
Extracting Information

- Annulus Decomposition
- Projection Unfolding
- Spectral Fitting
  - Obtain Temperature, Density, Abundances, Redshift
  - Derive Pressure, Entropy, Mass
- Spherically Symmetric Averaging
- Disturbed Systems Breakdown Possible



Deprojection





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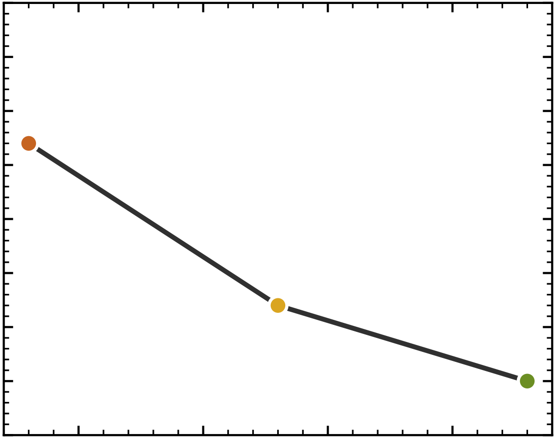

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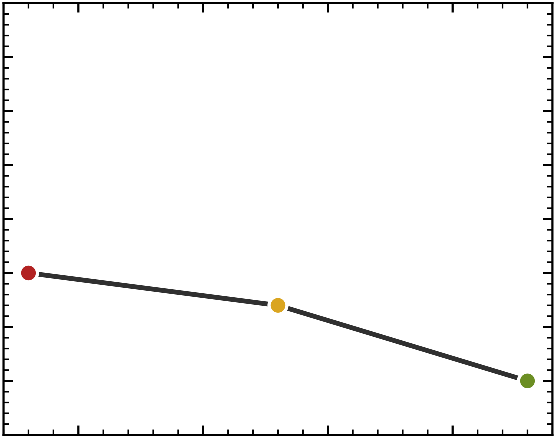
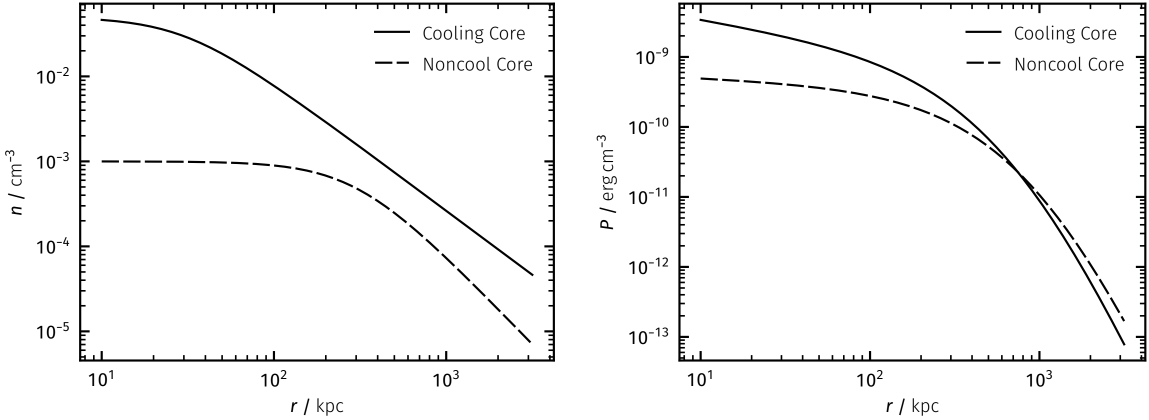


Overview

Density & Pressure



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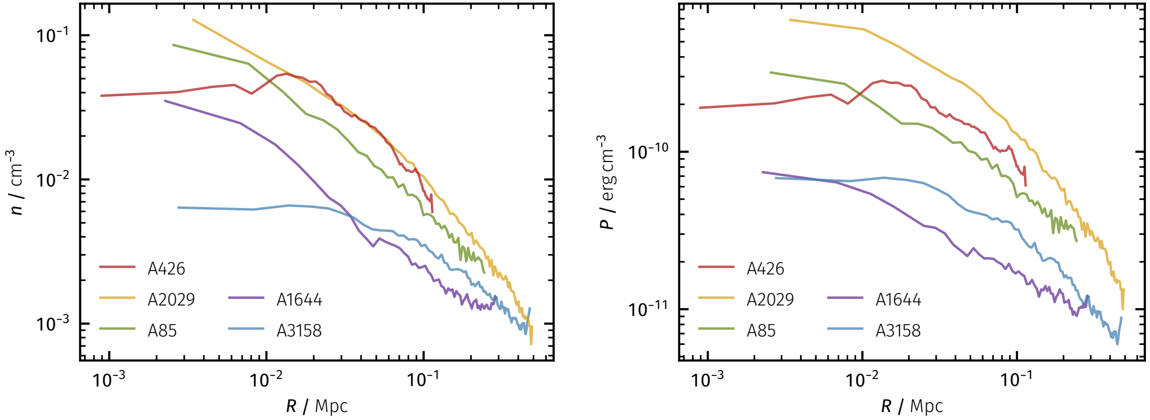

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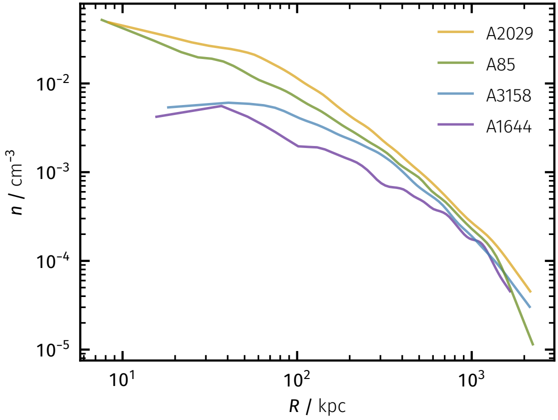
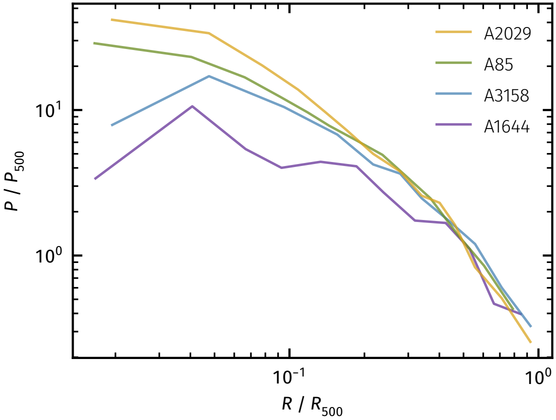
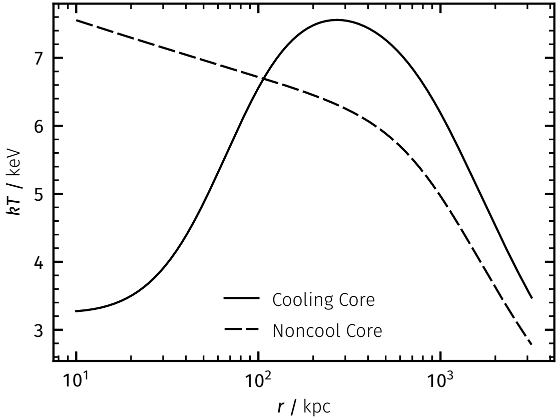
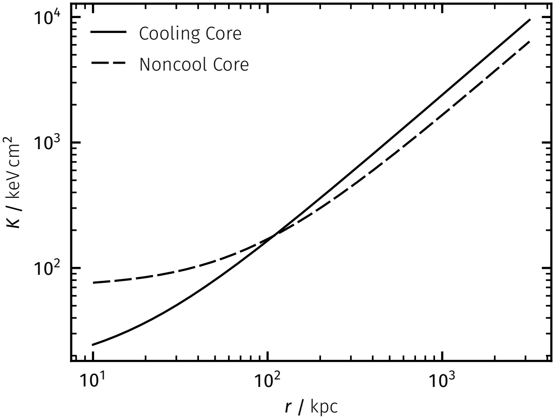




Temperature & Entropy





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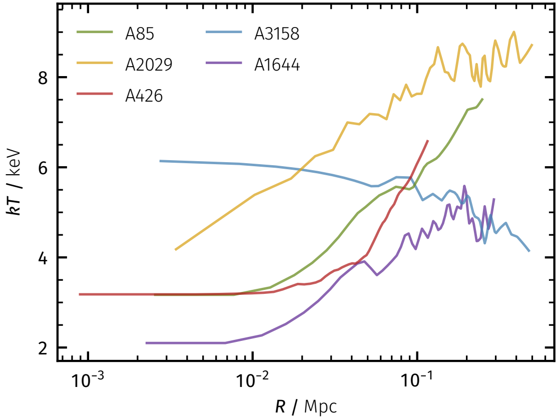

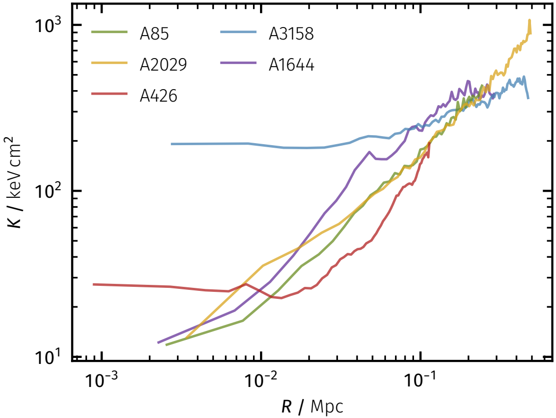

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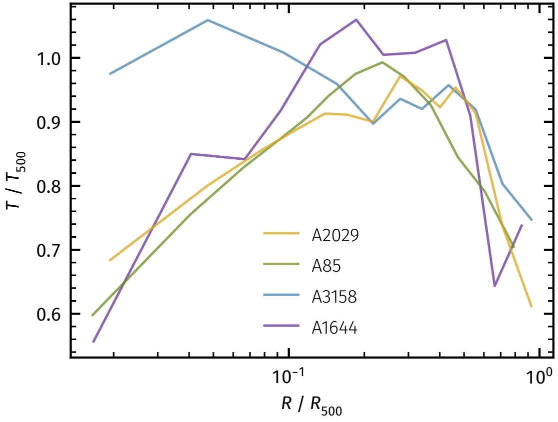
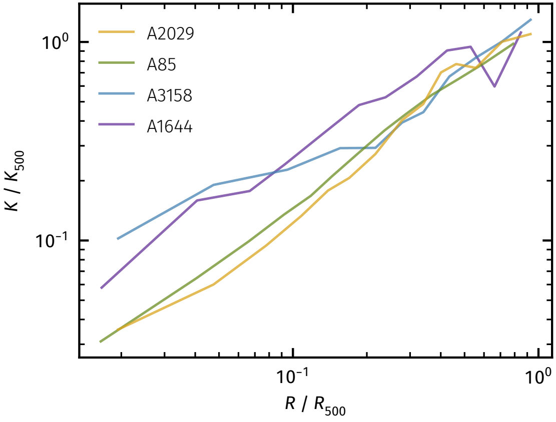




Overview

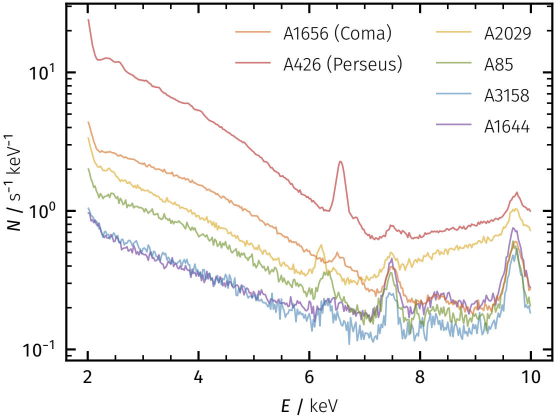
Selected Missions

- Large Survey
  - ROSAT *(retired)*
  - Planck *(retired)*
  - eROSITA *(active)*
- Spatial Resolution
  - Chandra *(active)*
  - NewAthena *(planned)*
- Spectroscopic Precision
  - ASCA *(retired)*
  - XMM-Newton *(active)*
  - XRISM *(active)*

Overview

Spectrum Analysis

- Continuum
  - Bremsstrahlung Baseline
  - Possible Soft & Hard Excesses
    - Multiphase Intracluster Gas
    - Warm-Hot Intergalactic Medium
    - Background Upscattering
- Line Emission
  - Powerful Flux Ratio Analyses
  - Fe K & L Complexes – Thermometers
  - O & Ne Lines – Supernova Tracers



Overview

Discussion

- X-Ray Spectroscopy Enables Cosmic Structure Evolution Inference
- CC & NCC Classification Framework Traces Cluster Thermodynamical History
- Starforming Galactic Wind & AGN Feedback Stabilize Cluster Cores Against Collapse
- Highly Resolved Simulations Promise Improved Dynamics Understanding
- Future Missions Probe ICM Microproperties & WHIM Filaments
- New Simulation & Data Turbulence Comparison

Thank You!

- H. Böhringer & N. Werner, *X-Ray Spectroscopy of Galaxy Clusters: Studying Astrophysical Processes in the Largest Celestial Laboratories* (2010) The Astronomy & Astrophysics Review
- K. W. Cavagnolo et al., *Intracluster Medium Entropy Profiles for a Chandra Archival Sample of Galaxy Clusters* (2009) The Astrophysical Journal Supplement
- D. Eckert et al., *The XMM Cluster Outskirts Project* (2017) Astronomische Nachrichten

Resources on GitHub: here

Appendix

- Total Contained Thermal Energy

$$E_t \propto nT$$

- Radiative Cooling Rate

$$\dot{E}_c \propto n^2 \Lambda_c(T) \propto n^2 T^{-1/2}$$

- Characteristic Cooling Time

$$t_c \equiv \frac{E_t}{\dot{E}_c} \propto T^{1/2} n^{-1}$$

- Bremsstrahlung Emissivity

$$\epsilon \propto n_e n_i Z^2 T^{-1/2} g_{ff}(T, E)\, e^{-E/kT}$$

Appendix

- Salpeter Initial Stellar Mass Function

$$\xi(M_*) = \frac{dN_*}{dM_*} \propto M_*^{-\alpha}$$

- Hydrostatic Equilibrium Condition

$$\nabla P = g\rho = -\frac{GM\rho}{r^2}$$

- Pressure Relation

$$P = nkT$$

- Entropy Relation

$$K = \frac{kT}{n^{2/3}}$$

Appendix

- Density Profile

$$n_{NCC} = n_0 \left(1 + \frac{r^2}{r_c^2}\right)^{-3\beta/2} \qquad n_{CC} = \sqrt{n_1^2 + n_2^2}$$

- Temperature Profile

$$T = T_0 \frac{x_{\rm cool} + T_{\rm min}/T_0}{x_{\rm cool} + 1} \frac{(r/r_t)^{-a}}{\left(1 + (r/r_t)^b\right)^{c/b}} \qquad x_{\rm cool} = (r/r_{\rm cool})^{a_{\rm cool}}$$

- Pressure Profile

$$P = P_0 \left((C_\ell\, x_\ell)^c \left(1 + (C_\ell\, x_\ell)^a\right)\right)^{-(b-c)/a}$$

- Entropy Profile

$$K = K_0 + K_\ell\, x_\ell^\gamma \qquad x_\ell = r/R_\ell$$